MEHTAB RIAZ

Full-Stack Engineer | Enterprise Applications | TypeScript · React · Node.js | Applied AI

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Borsbergstraße 34, Dresden, Saxony, Germany | Open to Hybrid / Relocation within Germany

Professional Summary

Full-Stack Engineer with **5+ years** building production-grade enterprise applications across fintech, crypto/blockchain analytics, and enterprise messaging — working across **front end and back end** in complex application landscapes. Strong in **TypeScript, JavaScript, React, Vue/Nuxt, Node.js, and REST/GraphQL API design**, with hands-on experience in **real-time systems, WebSockets, SSE, and large-scale data platforms**. Proven ability to **independently design and implement new features** in complex systems, optimize **usability and performance**, and produce **technical documentation**. Applies **AI pragmatically in day-to-day engineering** — LLM features in production, AI-assisted development workflows, and hybrid deterministic + AI decision systems. Experienced collaborating with **international, distributed teams** and communicating technical topics to non-technical stakeholders. Based in Dresden. *(German: A1, actively studying — see note below.)*

Technical Skills

Languages & Frameworks: TypeScript, JavaScript (ES2022+), React, Next.js, Vue 3, Nuxt 3 / Nuxt 4, Node.js, TanStack Start, Ruby on Rails, Python (familiar)

Front-End & UI: React, Redux Toolkit, RTK Query, TanStack Query/Router/Table/Form, React Hook Form, Zod, Tailwind CSS, SCSS, HTML5, Headless UI, shadcn/ui, Nuxt UI, accessible UI (semantic HTML, ARIA, keyboard navigation), Figma-to-code, Vite, Webpack

Back-End & APIs: REST API design, GraphQL, Express, Nitro, Node.js background jobs, synchronous and asynchronous API design, WebSocket / Server-Sent Events real-time APIs, webhook-driven integrations

AI in Engineering Practice: OpenAI API (GPT-4), LLM feature integration, natural-language-to-SQL pipelines, prompt engineering, AI agents and MCP orchestration, n8n orchestration, deterministic + AI hybrid decision systems, AI-assisted development (Cursor, Claude Code, Codex, sub-agents, custom skills)

Databases & Search: PostgreSQL, Redis, MongoDB, MySQL, Supabase, Elasticsearch, Typesense

Cloud & DevOps: AWS (Lambda, EC2, RDS, CloudWatch), Docker, Kubernetes (familiar), GitHub Actions, GitLab CI/CD, Jenkins, Terraform, Vercel, CI/CD automation

Testing: Jest, Vitest, Playwright, Cypress, RSpec, unit / integration / end-to-end testing

Identity, Payments & Compliance: Auth0 (SSO, MFA/TOTP, OAuth 2.0 / OIDC integration), JWT lifecycle, RBAC, Stripe API/SDK (PCI-conscious), GDPR / SOX / FINRA compliance, audit trails, data retention

Data Visualization: Highcharts, AG Grid, Chart.js, D3

Other: Git, npm, Agile/Scrum, technical documentation

Professional Experience

Senior Frontend Engineer — *Bridge by The Tie (Enterprise Communication Platform) — Strategic Systems International*

Aug 2025 – Present | Remote, Lahore → Dresden

Technologies: *Next.js, TanStack Start, React, React Native, Electron.js, TypeScript, WebRTC, WebSockets, SSE, Supabase Realtime, Auth0, OIDC, OAuth 2.0, MFA/TOTP, Stripe, PostHog, Global Relay, Cloudflare, Tailwind, React Hook Form, Zod, Redux Toolkit, RTK Query, TanStack Query/Router/Table/Form, Jest, Vitest, Playwright, Cypress, GitHub Actions, Docker*

- Architected and developed a **cross-platform enterprise messenger** after platform POCs — **Next.js** (web), **Electron** (macOS), and **React Native** (iOS/Android) — using Domain-Driven Design across **11 decoupled domain layers**, with a Supabase/PostgreSQL backend and strict TypeScript throughout.
- Independently designed and implemented new product features across front end and back end in a complex, multi-client application landscape — including real-time messaging (threading, reactions, read receipts, presence), rich media, and admin/compliance tooling.
- Implemented complete **WebRTC media stream handling** — audio capture, video transmission, call signaling, media device enumeration, stream constraints negotiation, ICE candidate handling, and STUN/TURN configuration — for reliable enterprise voice/video calling.
- Built real-time messaging and **synchronous/asynchronous APIs** with **WebSockets** (Supabase Realtime) and **Server-Sent Events (SSE)** — optimized for **1,000+ concurrent connections**.
- Integrated **Auth0** for SSO, social login, and MFA (TOTP) across web, mobile, and macOS clients; implemented application-side **OAuth 2.0 / OIDC** flows and **JWT** lifecycle, with **5-tier RBAC** and audit logging for SOX/FINRA/GDPR requirements.
- Designed and implemented an in-product **bot platform** — default Cloudflare bots with LLM access, plus admin-integrated external bots following Bridge's command rules, with custom commands under predefined rules; owned configuration, chat membership, command routing, and default-bot integration.
- Translated **Figma** designs into production React/Next.js components with **Tailwind CSS** using **Figma-to-code workflows** in Claude Code, Cursor, and Codex, guided by a custom agent skill enforcing consistent component structure.
- Delivered a full **enterprise compliance layer** (SOX, FINRA, GDPR) with audit trails and configurable data retention, plus an Admin Dashboard and Company Workspaces (profiles, team management, subscriptions, announcements).
- Integrated **Stripe** as the production payment gateway (Payment Intents, customer/payment-method APIs, webhook-driven status handling); instrumented **PostHog** analytics; integrated **Global Relay** for

communications archiving and insider-trading surveillance.
- Maintained **>85% test coverage** across WebRTC, auth, and messaging flows with Jest, Vitest, Playwright, and Cypress; designed and maintained **CI/CD** (GitHub Actions + Docker) with security scanning and compliance checks.

Senior Full-Stack Engineer (previously Software Engineer) — *Strategic Systems International (Institutional Financial Data Platform — The Tie Terminal)*

Jul 2022 – Present | Remote / Lahore, Pakistan

Technologies: *TypeScript, React, Vue 3, Nuxt 3 / Nuxt 4, Node.js, Nitro, Ruby on Rails, WebSockets, PostgreSQL, Elasticsearch, Typesense, AWS, OpenAI API, MCP, Highcharts, AG Grid, Stripe, PostHog, GitHub Actions, Docker, Figma, Claude Code, Cursor, Codex*

- Built a **natural-language-to-SQL pipeline** integrating the OpenAI API (GPT-4) for institutional analytics — feeding the live PostgreSQL schema as context, generating executable SQL, and surfacing tailored analytics results, cutting analyst query time from hours to seconds.
- Architected an **AI-assisted dashboard module** enabling non-technical users to compose custom widgets with dynamic Highcharts visualizations and AG Grid tables over live data streams — improving data discoverability and reducing time-to-insight.
- Owned and shipped the production **AI Mind-Map Canvas** (Vue 3 / Nuxt, TypeScript, Nuxt UI, Tailwind, Nitro): interactive, evidence-backed entity graphs across news, on-chain data, coins, companies, funds, and brands, with LLM/MCP-assisted graph commands.
- Led the **Vue 2 → Vue 3 → Nuxt 3 migration** on a high-traffic production platform — improving Time-to-Interactive by **~30%** and establishing company-wide TypeScript and Composition API standards.
- Implemented production **global search** with **Elasticsearch** and **Typesense** across news, companies, coins, funds, NFTs, datasets, data series, and charts; optimized **PostgreSQL, Elasticsearch, and Typesense** query and index performance through indexing, lazy loading, and caching — measurably improving Core Web Vitals on data-heavy views.
- Engineered a **multi-channel real-time alert system** (Discord, Slack, Telegram, email, in-app/WebSocket) using Node.js and Rails background workers, monitoring **1,000+ tracked crypto assets** with sub-100ms delivery latency.
- Shipped the **crypto screener module** — large tabular datasets for coins, companies, and funds enriched with integrated charts, real-time alerts, and related news — plus the platform's news/on-chain events section.
- Configured an **AI code reviewer** with custom security-pattern instructions that surfaced a critical authentication vulnerability (stale session handling) missed across multiple human review cycles.
- Drove **CI/CD** improvements via GitHub Actions and Docker; contributed to AWS deployment automation (Lambda, EC2, RDS, CloudWatch) and observability; mentored junior engineers, led architectural discussions on DDD/REST API design, and engaged directly with clients in requirements-gathering aligned to FINRA and GDPR.

Assistant Software Engineer — *Devsinc (SaaS Hotel Management Platform)*

Jan 2022 – Jul 2022 | Lahore, Pakistan

Technologies: *Node.js, Ruby on Rails, REST API, PostgreSQL, MySQL, MongoDB, Firebase, Stripe, RBAC, JWT, Python (Selenium)*

- Developed booking and **multi-role access control** features for a hotel management SaaS platform; implemented secure REST APIs with JWT authentication and RBAC, consumed by frontend and third-party integrations.
- Integrated **Stripe** as the production payment gateway for hotel-booking checkout (Payment Intents, customer/payment-method management, webhook-driven status handling, idempotent event processing).
- Refactored complex SQL queries across **PostgreSQL and MySQL**, improving data retrieval performance by **40%** through indexing and query restructuring.
- Built and maintained **MongoDB and Firebase** integrations; automated client data-collection pipelines using Python Selenium and MS Web Scraper.

Software Engineer — *Freelance*

Jan 2021 – Dec 2021 | Lahore, Pakistan (Remote)

- Built cross-platform **desktop applications** using Electron.js and Vue for local business clients, including barcode scanning and point-of-sale workflows.
- Trained and fine-tuned an **NLP model for Roman Urdu text classification**, handling dataset curation drawn from ~50 research publications.

Selected Projects

The Tie Terminal — Institutional Crypto Data Platform *TypeScript, Nuxt 3 / Nuxt 4, Vue 3, Node.js, Nitro, Ruby on Rails, PostgreSQL, Elasticsearch, Typesense, OpenAI API, MCP, AG Grid, AWS* Comprehensive data platform for institutions working with digital assets; shipped crypto screener for **1,000+ assets** with integrated charts, alerts, and news enrichment; built global search over news, companies, coins, funds, NFTs, datasets, data series, and charts with Elasticsearch and Typesense; owned and shipped the production AI Mind-Map Canvas; integrated Stripe and PostHog; built the natural-language-to-SQL LLM pipeline.

Frosty Metrics — Blockchain Analytics Platform *Jan 2025 – Present | Nuxt 3, Vue 3, TypeScript, Nitro, Highcharts | frostymetrics.com* Live analytics platform for the Avalanche blockchain — DeFi, NFT, Gaming, and Subnet engagement via interactive Highcharts dashboards for complex multi-series time-series data at scale. Designed the **Chill Factor scoring system** — a composite engagement metric synthesizing on-chain activity signals into a single developer-facing performance indicator.

AI Lead Qualification & Follow-Up System *n8n, OpenAI (structured output), PostgreSQL/Supabase, TypeScript, Slack, APIs/Webhooks, Next.js* Next.js application plus n8n/OpenAI backend that captures, validates, scores, and routes inbound sales leads — replacing manual triage with a **hybrid deterministic + AI decision engine**. Transparent 100-point scoring model across six weighted dimensions, with duplicate detection, data-completeness, risk, and urgency gates, keeping the LLM out of final score determination for explainability. Idempotent webhook handling, entity resolution, retry handling, and full audit trail.

Real-Time Collaborative Editor — Open Source *2023 | React, Node.js, MongoDB, WebSockets, Yjs (CRDT) | github.com/MehtabRiaz/Real-time-Collaborative-Editor* MERN-stack real-time collaborative editor using Yjs CRDT for conflict-free multi-user editing, WebSocket presence indicators, and document-level access control.

Education

University of the Punjab (FCIT) — BS Computer Science *2018 – 2022 | Lahore, Pakistan | GPA: 3.63 / 4.0 (German Grading Equivalent: 1.6)*

Certifications

- **Claude Academy: AI Fluency: Framework & Foundations** — Anthropic (Issued Sep 2026).
- **Certificate of Participation, Codegoda 2023** — Unstop (Issued Apr 2023).

Languages

English (C1 — Professional Proficiency), Urdu (Native), **German (A1 — actively studying)**